

Multiple Choice

1. **Answer:** C

Options: A) 0.3; B) 0.7; C) 0.15; D) 0.9; E) 0.5

Note: Correct option: C - 0.15

2. **Answer:** E

Options: A) 1.5×10^{-4} grams; B) 2.50grams; C) 3.33grams; D) 4.00grams; E) 1.00grams

Note: Correct option : E – 1.00grams

2. **Answer:** B

Options: A) Redox reactions; B) Elementary processes; C) Dissolution reactions; D) Polymerization reactions; E) Complexation reactions

Note: Correct option: B - Elementary processes

3. **Answer:** A

Options: A) CH₄; B) NH₃; C) CCl₂F₂; D) H₂O; E) SF₆

Note: Correct option: A - CH₄

4. **Answer:** A

Options: A) 1; B) 0.75; C) 2; D) 4; E) 0.5

Note: Correct option: A - 1

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '25.3 days'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2.5 g'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'bromine'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'pH is 4, pOH is 4'.

Long Form Response

11. **Answer:** 72.42% and 27.58%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 0.273 g. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.